SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Cover Table

Field	Value
Project	Department Course Timetable Management System
Document	Software Requirements Specification
Version	v2
Date	10.11.2026
Team number	[Team 21]
Member 1	[Buğrahan KOÇASLAN] · [admbugraa]
Member 2	[Furkan KÖRPE] · [Furkankrp]
Repository	https://github.com/Furknkrp/-Timetable-Management-System

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Revision History

Version	Date	Author(s)	Summary of changes
v1	1.11.2026	Buğrahan KOÇASLAN, Furkan KÖRPE	Initial requirements from the project description
v2	10.11.2026	Buğrahan KOÇASLAN, Furkan KÖRPE	Updated for Code v1/v2 milestone: added violation messages, input error handling details, and screen mappings

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1. Introduction

1.1 Purpose

This document specifies the software requirements for the Department Course Timetable Management System. It serves as the primary reference for the development team and the course coordinator to understand system functionalities, constraints, and operational boundaries.

1.2 Scope

- **In scope:** Loading course, instructor, room, and constraint data from external text/CSV files; automatic weekly timetable generation for year groups 1 to 4; conflict detection and warning mechanisms; manual session modification; generation of classroom request email drafts.
- **Out of scope:** Database integration, graphical web/desktop user interface, automated email sending over SMTP, and student-level individual elective course enrollment conflict resolution.

1.3 Definitions

Term	Definition
Session (block)	One theory (`-T`) or laboratory (`-L`) block of a course, placed in consecutive one-hour slots.
Time slot	A one-hour scheduling unit between 09:00 and 17:00 on weekdays.
Year group	The academic year of undergraduate students (1st, 2nd, 3rd, or 4th year).
Hard constraint	A mandatory scheduling rule that must never be violated under any circumstances.
Soft constraint	A preference rule; violation does not prevent scheduling but generates a warning.
Special request	Course-specific predefined requirements such as fixed times or specific rooms.
Unscheduled session	A session that could not be placed into the timetable due to room shortages or constraint conflicts.

1.4 References

- Term Project Assignment: Department Course Timetable Management System
- SENG 271 Project Guidelines and Template Specifications

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2. Overall Description

2.1 Product Perspective

A stand-alone Java console application that reads configuration and data files from the `data/` directory, processes scheduling logic via rule-based checks, displays menu-driven screens, and exports weekly timetables and reports to the `output/` directory.

2.2 User

User	Main goals	Characteristics
Department Course Timetable Coordinator	Load data, generate timetables, inspect conflicts, modify slots manually, and export reports.	Department Course Timetable Coordinator with basic command-line navigation skills.

2.3 Operating Environment

Java SE Development Kit (JDK 17 or higher), cross-platform operating system (Windows/macOS/Linux), executed via console terminal or IDE.

2.4 Assumptions and Constraints

- Working days are Monday through Friday, 09:00 to 17:00, structured in one-hour slots.
- Each course belongs to exactly one year group.
- Theory and laboratory hours are treated as separate blocks and are never split across non-consecutive slots.
- External input files conform to expected CSV and TXT syntax.

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3. Functional Requirements

ID	Group	Requirement	Priority	Source	Screen
FR-01	Data loading	The system shall read course data from a CSV file.	Must	§3.7	Screen 2
FR-02	Data loading	The system shall skip a record that contains an invalid value and show a warning, without terminating.	Must	§3.7	Screen 2
FR-03	Data management	The system shall parse and store instructor availabilities, room capacities, and constraints from external files.	Must	§3.1–3.4	Screen 2
FR-04	Automatic generation	The system shall automatically schedule fixed-time courses first and then assign remaining sessions using a first-fit algorithm.	Must	§3.8	Screen 3
FR-05	Conflict handling	The system shall record unassigned sessions in an unscheduled list along with the exact failure reason (`NO ROOM` or `CONSTRAINT`).	Must	§3.5, §3.8	Screen 5
FR-06	Viewing and modification	The system shall display weekly schedules per year group and allow manual session adjustments under hard constraint validation.	Must	§3.9	Screens 4, 6
FR-07	Classroom request	The system shall generate a formal classroom request email text file for sessions left unassigned due to room shortages.	Must	§3.10	Screen 7
FR-08	Output	The system shall export final timetables and operational	Must	§3.11	—

	Group	Requirement	Priority	Source	Screen
		summaries into the `output/` directory.			

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4. Scheduling Constraints

4.1 Hard Constraints (must never be violated)

ID	Constraint	Rule in our own words	Data needed to check it	Message on violation (v2)
HC-01	Instructor conflict	An instructor cannot teach two sessions simultaneously.	Instructor ID, Day, TimeSlot	`[REJECTED] Instructor is already busy at this time.`
HC-02	Room conflict	A room cannot host multiple sessions at the same time.	Room ID, Day, TimeSlot	`[REJECTED] Room is already occupied.`
HC-03	Year-group conflict	Mandatory courses of the same year group cannot overlap.	Year Group, Day, TimeSlot	`[REJECTED] Year group has a conflicting mandatory course.`
HC-04	Working hours	Sessions must fit entirely between 09:00 and 17:00.	StartHour, Duration, EndHour	`[REJECTED] Session exceeds working hours (09:00-17:00).`
HC-05	Room capacity	Room capacity must meet or exceed student count.	Room Capacity, Student Count	`[REJECTED] Room capacity is insufficient for student count.`
HC-06	Room type	Theory sessions require classrooms; labs require laboratories.	Session Type, Room Type	`[REJECTED] Room type does not match session requirement.`
HC-07	Instructor availability	Instructors cannot be assigned during unavailable slots.	Instructor ID, UnavailableSlots	`[REJECTED] Instructor is unavailable during this time slot.`
HC-08	Fixed-time requirements	Predefined fixed-time requests must be respected.	Course Code, FixedSlot	`[REJECTED] Fixed-time constraint violation.`

ID	Constraint	Rule in our own words	Data needed to check it	Message on violation (v2)
HC-09	Room availability	Rooms cannot be assigned during maintenance/unavailable slots.	Room ID, UnavailableSlots	`[REJECTED] Room is unavailable during this time slot.`

4.2 Soft Constraints (preferences; a violation produces a warning)

ID	Constraint	How our system considers it	Warning shown
SC-01	Theory before laboratory	Checks if theory precedes lab; warns if reversed.	`[WARNING] Lab session is scheduled before theory.`
SC-02	Instructor preferences	Checks if assigned slot matches instructor preferences.	`[WARNING] Session placed outside instructor preferred time.`
SC-03	Avoiding unnecessary gaps	Detects idle hours between daily sessions for a year group.	`[WARNING] Unnecessary time gap created in year schedule.`
SC-04	Consecutive teaching hours	Ensures multi-hour blocks are scheduled consecutively.	`[WARNING] Teaching block split or non-consecutive.`

4.3 Special Scheduling Requests

Request type	Example	Mandatory or preferred	How the system handles it
Fixed time	SENG301 must be on Wednesday, 13:00–16:00	Mandatory (Hard)	Processed in Phase 1 of the scheduling algorithm.
Specific room	SENG401 must use Laboratory B202	Mandatory / Preferred	Validated against room availability and capacity.
Instructor unavailable day	SENG201 instructor unavailable on Friday	Mandatory (Hard)	Blocked during constraint checking (HC-07).
Laboratory before theory	SENG302 laboratory may precede theory	Preferred (Soft)	Bypasses SC-01 warning when explicitly requested.

5. Input Data Requirements

File	Format	Fields (in order)	Example line	Validation rules
data/courses.csv	CSV	code, title, year, instructorId, theoryHours, labHours, students, roomType	`SENG101,Intro to SE,1,INS1,3,2,60,Classroom`	year is 1–4; hours are whole numbers; instructor must exist
data/instructors.txt	TXT	instructorId, name, unavailableSlots	`INS1,Dr. John Doe,FRI:15:00-17:00`	ID must be unique; format must match day/time pattern.
data/rooms.csv	CSV	roomId, roomType, capacity	`B202,Laboratory,50`	capacity > 0; type must be Classroom or Laboratory.
data/constraints.txt	TXT	courseCode, requestType, details	`SENG301,FIXED,WED:13:00-16:00`	Course code must exist in courses.csv.

Input error handling

Error	Example	System reaction
Missing file	`rooms.csv` not found	Displays file error message, prompts user to check `data/` folder without crashing.
Invalid numeric value	`"three"` for theory hours	Skips the malformed line, logs warning, and continues loading.
Incomplete record	A course line with a missing field	Discards record, outputs warning line number.
Reference to nonexistent item	Request for course SENG499	Ignores request rule and reports warning.

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6. Output Requirements

Output	Content	Shown on screen	Saved to file
Weekly timetable per year group	Day, time, course code and title, instructor, room	Yes	Optional export

Output	Content	Shown on screen	Saved to file
All scheduled sessions	Comprehensive list of active assignments	Yes	Yes
Unscheduled sessions	Session, problem type (no room / constraint), reason	Yes	Yes
Warnings	Soft-constraint and input warnings	Yes	Yes
Classroom request email draft	Course, number of students, room type, requested time	Yes	`output/classroom_request_<Code>.txt`

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7. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Reliability	The system shall not terminate unexpectedly on invalid input; it shall show an understandable message instead.
NFR-02	Usability	The system shall reject an invalid menu choice with a message and show the menu again.
NFR-03	Performance	Timetable generation for up to 50 courses shall complete in under 3 seconds.
NFR-04	Maintainability	Source code shall follow standard Java object-oriented encapsulation and naming conventions.

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8. Changes from the Previous Version

ID or section	Change (added / changed / removed)	Reason for the change
Section 4.1	Added violation messages for HC-01 through HC-09	To support v2 interactive constraint verification and console feedback.
Section 5	Detailed CSV/TXT file fields and error recovery rules	Aligned with Code v1 implementation requirements.

AI-Use Declaration

Tool	Used for	How we checked the result
Gemini	**Structuring SDD Sections & Content Formatting:** Assisting in organizing document sections according to project template guidelines, generating Mermaid syntax for class and flowchart diagrams, and formatting memory model / traceability tables.	Both team members thoroughly reviewed all generated section structures to match SENG 271 course specifications.